בס" ד

TLS Class Exercise (Barak Gonen):

I. TLS v1.2 handshake

Open the snipping file **TLS12sniffEx.pcapng**.

1. How many packets contain Client Hello? What is the filter that allows you to find packets of this type only? 42 packets. Filter - tls.handshake.type == 1
2. Focus on the first Client Hello packet. What is the domain name to which you are browsing to?

```
                Cipher Suites (17 suites)
           Compression Methods Length: 1
          Compression Methods (1 method)
                  Extensions Length: 401
        Extension: server_name (len=21)
               Type: server_name (0)
                          Length: 21
    Server Name Indication extension
      Server Name list length: 19
  Server Name Type: host_name (0)
           Server Name length: 16
     Server Name: www.hattrick.org
```

3. What filter can be put in Wireshark to receive only the Client Hello, which is intended for a Server Name of our choice? tls.handshake.extensions_server_name == www.hattrick.org

   tcp.stream eq 9 - filter
4. What is the TLS version?

Info	Length	Protocol	Destination	Source	Time	.I
Client Hello	571	TLSv1.2	193.34.188.81	192.168.1.221	3.112884	66

5. What is the Cipher Suite (קבוצת צפנים) chosen by the server? What algorithm is used for the following operations?

   I right-clicked on the client hello - **follow tcp stream** entry and accepted filtering for the conversation between the client and the server. I looked for a **server hello packet** and there I saw the Cipher Suite that the server had chosen.

```
Session ID Length: 32
Session ID: e6250000315d96db78bef8014ba84fa6c19ce27aa47353ea…
Cipher Suite: TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384 (0xc030)
```

   a. Hashing - SHA-384

   b. Encryption – AES-256-GCM

   c. Authentication - RSA

   d. Key Exchange - ECDHE
6. How many certificates are sent by the server? What is their common name?

   2 certificates

```
Certificates (3439 bytes)
    Certificate Length: 1946
  > Certificate: 308207963082057ea003020102021004f474c32e71498906… (id-at-commonName=*.hattrick.org)
    Certificate Length: 1487
  > Certificate: 308205cb308204b3a003020102021 00f622f6f21c2ff5d52… (id-at-commonName=GeoTrust Global TLS RSA4096 SHA256 2022 CA1,id-at-organizationName=D…
```

7. Is the certificate also valid for other URLs that are under hattrick.org? yes, If *.hattrick.org appears, the certificate is a Wildcard and is valid for all subdomains under the main domain.

8. Between which dates is the hattrick.org certificate valid?

```
validity
  notBefore: utcTime (0)
      utcTime: 22-06-30 00:00:00 (UTC)
  notAfter: utcTime (0)
      utcTime: 23-07-15 23:59:59 (UTC)
```

9. Who is the Root Certificate Authority and was his certificate sent to the client?

The Root Certificate Authority does not appear because it already appears in the browser and is known to everyone.

```
Certificates (3439 bytes)
    Certificate Length: 1946
  > Certificate: 308207963082057ea003020102021004f474c32e71498906… (id-at-commonName=*.hattrick.org)
    Certificate Length: 1487
  Certificate: 308205cb308204b3a003020102021 00f622f6f21c2ff5d52… (id-at-commonName=GeoTrust Global TLS RSA4096 SHA256 2022 CA1,id-at-organizationName=D…
    signedCertificate
        version: v3 (2)
        serialNumber: 0x0f622f6f21c2ff5d521f723a1d47d62d
      > signature (sha256WithRSAEncryption)
      issuer: rdnSequence (0)
        > rdnSequence: 4 items (id-at-commonName=DigiCert Global Root CA,id-at-organizationalUnitName=www.digicert.com,id-at-organizationName=DigiCert …
      > validity
```

10. How does the customer verify that the hattrick.org certificate is not revoked? Is it through the Certificate Revocation List, through the OCSP protocol (Online Certification Status Protocol) or through OCSP Stapling? 1. Certificate Revocation List 2. OCSP

```
Certificates (3439 bytes)
    Certificate Length: 1946
  Certificate: 308207963082057ea003020102021004f474c32e71498906… (id-at-commonName=*.hattrick.org)
    signedCertificate
        version: v3 (2)
        serialNumber: 0x04f474c32e71498906b6dc88c8763dae
      > signature (sha256WithRSAEncryption)
      > issuer: rdnSequence (0)
      > validity
      > subject: rdnSequence (0)
      > subjectPublicKeyInfo
      extensions: 10 items
        > Extension (id-ce-authorityKeyIdentifier)
        > Extension (id-ce-subjectKeyIdentifier)
        > Extension (id-ce-subjectAltName)
        > Extension (id-ce-keyUsage)
        > Extension (id-ce-extKeyUsage)
        Extension (id-ce-cRLDistributionPoints)
            Extension Id: 2.5.29.31 (id-ce-cRLDistributionPoints)
          CRLDistPointsSyntax: 2 items
```

```
        Extension Id: 2.5.29.31 (id-ce-cRLDistributionPoints)
      CRLDistPointsSyntax: 2 items
        DistributionPoint
          distributionPoint: fullName (0)
            fullName: 1 item
              GeneralName: uniformResourceIdentifier (6)
                uniformResourceIdentifier: http://crl3.digicert.com/GeoTrustGlobalTLSRSA4096SHA2562022CA1.crl
        > DistributionPoint
    > Extension (id-ce-certificatePolicies)
    Extension (id-pe-authorityInfoAccess)
        Extension Id: 1.3.6.1.5.5.7.1.1 (id-pe-authorityInfoAccess)
      AuthorityInfoAccessSyntax: 2 items
        AccessDescription
            accessMethod: 1.3.6.1.5.5.7.48.1 (id-ad-ocsp)
          accessLocation: 6
              uniformResourceIdentifier: http://ocsp.digicert.com
        > AccessDescription
```

11. Which Record is sent from the server to the client and would not have been sent if the key exchange was with the RSA algorithm? What does this record include when it is sent?

In the server hello package, click on it -> click on key exchange -> there is a Pubkey (= public key) under DH algorithms. The record includes when sent: Pubkey, Named Curve

```
> Handshake Protocol: Server Hello
> Handshake Protocol: Certificate
v Handshake Protocol: Server Key Exchange
    Handshake Type: Server Key Exchange (12)
    Length: 296
  v EC Diffie-Hellman Server Params
      Curve Type: named_curve (0x03)
      Named Curve: x25519 (0x001d)
      Pubkey Length: 32
      Pubkey: 0524cd7f90ce7b87cafa67994443f90f261bc104aecd5db4…
    > Signature Algorithm: rsa_pkcs1_sha256 (0x0401)
      Signature Length: 256
      Signature: 899dc8db1b5e58545f5f804a312134d915487546b31de71a…
```

12. What is the handshake type of the Client Key Exchange and what does it include? Does the customer sign it? Click on the client key exchange -> key exchange -> DH:

ניתן לראות שהלקוח מעביר את ה PUBKEY שלו, The client does not sign it.

```
> Transmission Control Protocol, Src Port: 59047, Dst Port: 443, Seq: 518
v Transport Layer Security
  v TLSv1.2 Record Layer: Handshake Protocol: Client Key Exchange
      Content Type: Handshake (22)
      Version: TLS 1.2 (0x0303)
      Length: 37
    v Handshake Protocol: Client Key Exchange
        Handshake Type: Client Key Exchange (16)
        Length: 33
      v EC Diffie-Hellman Client Params
          Pubkey Length: 32
          Pubkey: e440e2678bd4a3619447555de73ba6ff8d3d7ddb66958da7…
```

13. Use the SSL log file to decrypt the encrypted information. How can wireshark pick a specific set of keys, from all the keys in the file, which is the correct key set for the TLS session? There is a value which appears both in the sniffing and in the Wireshark log. What is the name of that field, in wireshark? Click on the client hello packet -> inside the handshake protocol there is a random field. The content of random appears in the SSL file and in the same line appears the encryption key that is used to decrypt all transport.

```
> Frame 66: 571 bytes on wire (4568 bits), 571 bytes captured (4568 bits) on interfac
> Ethernet II, Src: bc:0f:9a:2b:90:93 (bc:0f:9a:2b:90:93), Dst: Technico_22:47:7e (d4
> Internet Protocol Version 4, Src: 192.168.1.221, Dst: 193.34.188.81
> Transmission Control Protocol, Src Port: 59047, Dst Port: 443, Seq: 1, Ack: 1, Len:
v Transport Layer Security
  v TLSv1.2 Record Layer: Handshake Protocol: Client Hello
      Content Type: Handshake (22)
      Version: TLS 1.0 (0x0301)
      Length: 512
    v Handshake Protocol: Client Hello
        Handshake Type: Client Hello (1)
        Length: 508
        Version: TLS 1.2 (0x0303)
      > Random: b9bdf1866b10cea7dac8676e9234afaf1a4898c9dfffa5c1…
        Session ID Length: 32
```

```
CLIENT_RANDOM af487cb9b9cdf7b0d713c61725ca4e9455816a751e1e27682dd371b638586114 554bd779449949c6a2c83736bd685373e8a22537d7
CLIENT_RANDOM 6de1b8132f488155fcc7445fa3a521ab9750b8fa1dd91324fec0ee26a4db7a68 92dd09dd88e0ba20f31973b4ccabbfbdbc777a73b49
CLIENT_RANDOM b9bdf1866b10cea7dac8676e9234afaf1a4898c9dfffa5c174aeb75a77599d22 d56e64ec4c85711dac8ced1b6cdfef79f0ee95ba1bb
CLIENT_RANDOM 789d8728783b42dd7ef6c012d0d44301d40580d0e187cb020a0de570a6a3aa5 e5bb19464dae61f00fed21132c9ff2562e669056d81
```

14. What is the first resource that the client does a GET for? What is the received status code?

Explain what the customer will do as a result. We will look for the GET / packet and after it a packet appears that was sent following the GET called moved permanently. We click on it -> Then the client does GET / en-us the resource that appeared there

moved permanently

```
HyperText Transfer Protocol 2
  Stream: HEADERS, Stream ID: 15, Length 479, 301 Moved Permanently
      Length: 479
      Type: HEADERS (1)
    > Flags: 0x04, End Headers
      0... .... .... .... .... .... .... .... = Reserved: 0x0
      .000 0000 0000 0000 0000 0000 0000 1111 = Stream Identifier: 15
      [Pad Length: 0]
      Header Block Fragment […]: 3fe11f48826401588daec3771a4bf4a523f2b0e62c005f92497ca589
      [Header Length: 819]
      [Header Count: 15]
    > Header table size update
    > Header: :status: 301 Moved Permanently
    > Header: cache-control: private, max-age=0
    > Header: content-type: text/html; charset=utf-8
    > Header: location: https://www.hattrick.org/en-us/
```

GET / EN-US

```
      Length: 171
      Type: HEADERS (1)
    > Flags: 0x25, Priority, End Headers, End Stream
      0... .... .... .... .... .... .... .... = Reserved: 0x0
      .000 0000 0000 0000 0000 0000 0001 0001 = Stream Identifier: 17
      [Pad Length: 0]
      0... .... .... .... .... .... .... .... = Exclusive: False
      .000 0000 0000 0000 0000 0000 0000 1101 = Stream Dependency: 13
      Weight: 41
      [Weight real: 42]
      Header Block Fragment […]: 82058560b52d6a18c987c8c7c6c5c460a18776b5f4e0
      [Header Length: 761]
      [Header Count: 17]
    > Header: :method: GET
    > Header: :path: /en-us/
    > Header: :authority: www.hattrick.org
```

15. Follow TLS Stream. In the search screen, search for **Username** and **Password**.

We click on the FOLLOW -> TLS STREAM -> tab, we see the HTML of the site and we do a username search and look for what is colored red (because that is what the client sent)

```
ADfg0jBzdDbq8Bm52aW5fUlDrnzeZwGL2LbvXX6wjG%2BXPHzwY5aqHPjQceluVkcYznGym4jsDyiCJlx%2Bq6L0BUzor3I1sJx
24ucMenu%24ucLanguages%24ddlLanguages=151&loginname=&password=&loginname=%7B%7Bsignup.loginname%
in%24txtUserName=wireshark1&ctl00%24CPContent%24ucLogin%24txtPassword=TLSrulez
.....
N..4vsm..5.........-          ..SyQG...G..g.a&.ii....K5h..~....s.........E.f.......X(....v...D..Xow....*.....5.tw.....x.*)op.6\.*.....{...".....}
...@.mDJ.}.......1h...cL.1..
Kl/.....UG..7...2K.q.....a..Y>....D. J.....S...\.528.........<html><head><title>Object moved</title></head><body>
<h2>Object moved to <a href="//www89.hattrick.org/?authCode=2iHQj%2b%2bIG4lexRWCplE0eYz6d0fFr8jpgker%2fDh
Packet 2332. 63 client pkts, 116 server pkts, 32 turns. Click to select.
Entire conversation (632 kB)        Show as ASCII        No delta times
Find: UserName
```

II. TLS v1.3 Handshake

Open the snipping file **TLS13sniffEx.pcapng**. Find the stream associated with **jct.ac.il**.

Filter = frame contains "jct.ac.il" -> click follow TLS stream [= filter - tcp.stream eq 17]

frame contains "jct.ac.il"

No.	Time	Source	Destination	Protocol	Length	Info
652	7.200176	192.168.1.174	185.186.66.220	TLSv1.3	409	Client Hello

Client Hello:

16. What version exists in the Record Layer Header? What version exists in the Client Hello Version field? Version TLS 1.2

```
Transport Layer Security
  TLSv1.3 Record Layer: Handshake Protocol: Client Hello
    Content Type: Handshake (22)
    Version: TLS 1.0 (0x0301)
    Length: 1810
    Handshake Protocol: Client Hello
      Handshake Type: Client Hello (1)
      Length: 1806
      Version: TLS 1.2 (0x0303)
```

17. What are the first 6 hex digits of the Client Random number?

```
Transport Layer Security
  TLSv1.3 Record Layer: Handshake Protocol: Client Hello
    Content Type: Handshake (22)
    Version: TLS 1.0 (0x0301)
    Length: 1810
    Handshake Protocol: Client Hello
      Handshake Type: Client Hello (1)
      Length: 1806
      Version: TLS 1.2 (0x0303)
      Random: 38d4f6af7dd9576cb9f15dc2ae54d6187eb3b9553a544d26…
      Session ID Length: 32
      Session ID: 39d56ae08caa00402e8e73c52857faa0ac525e2be0e4cf70…
      Cipher Suites Length: 32
      Cipher Suites (16 suites)
```

18. What are the first 6 hex digits of the Session ID?

```
Transport Layer Security
  TLSv1.3 Record Layer: Handshake Protocol: Client Hello
    Content Type: Handshake (22)
    Version: TLS 1.0 (0x0301)
    Length: 1810
    Handshake Protocol: Client Hello
      Handshake Type: Client Hello (1)
      Length: 1806
      Version: TLS 1.2 (0x0303)
      Random: 38d4f6af7dd9576cb9f15dc2ae54d6187eb3b9553a544d26…
      Session ID Length: 32
      Session ID: 39d56ae08caa00402e8e73c52857faa0ac525e2be0e4cf70…
      Cipher Suites Length: 32
      Cipher Suites (16 suites)
```

19. How many Cipher Suites did the client propose?

```
Transport Layer Security
  TLSv1.3 Record Layer: Handshake Protocol: Client Hello
    Content Type: Handshake (22)
    Version: TLS 1.0 (0x0301)
    Length: 1810
    Handshake Protocol: Client Hello
      Handshake Type: Client Hello (1)
      Length: 1806
      Version: TLS 1.2 (0x0303)
      Random: 38d4f6af7dd9576cb9f15dc2ae54d6187eb3b9553a544d26…
      Session ID Length: 32
      Session ID: 39d56ae08caa00402e8e73c52857faa0ac525e2be0e4cf70…
      Cipher Suites Length: 32
      Cipher Suites (16 suites)
```

20. Which Cipher Suites are TLS 1.3?

Cipher Suites with "with" in the middle = TLS 1.2

Cipher Suites without "with" in the middle = TLS 1.3

```
Cipher Suites (16 suites)
    Cipher Suite: Reserved (GREASE) (0x9a9a)
    Cipher Suite: TLS_AES_128_GCM_SHA256 (0x1301)
    Cipher Suite: TLS_AES_256_GCM_SHA384 (0x1302)
    Cipher Suite: TLS_CHACHA20_POLY1305_SHA256 (0x1303)
    Cipher Suite: TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256 (0xc02b)
    Cipher Suite: TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 (0xc02f)
    Cipher Suite: TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384 (0xc02c)
    Cipher Suite: TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384 (0xc030)
    Cipher Suite: TLS_ECDHE_ECDSA_WITH_CHACHA20_POLY1305_SHA256 (0xcca9)
    Cipher Suite: TLS_ECDHE_RSA_WITH_CHACHA20_POLY1305_SHA256 (0xcca8)
    Cipher Suite: TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA (0xc013)
    Cipher Suite: TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA (0xc014)
    Cipher Suite: TLS_RSA_WITH_AES_128_GCM_SHA256 (0x009c)
    Cipher Suite: TLS_RSA_WITH_AES_256_GCM_SHA384 (0x009d)
    Cipher Suite: TLS_RSA_WITH_AES_128_CBC_SHA (0x002f)
    Cipher Suite: TLS_RSA_WITH_AES_256_CBC_SHA (0x0035)
```

21. Use the Extension “Key_Share”. Which Diffie-Hellman curves are chosen by the client?

Extension key share -> The client supports 3 Diffie-Hellman curves

```
Extension: key_share (len=1263)
    Type: key_share (51)
    Length: 1263
    Key Share extension
      Client Key Share Length: 1261
      Key Share Entry: Group: Reserved (GREASE), Key Exchange length: 1
      Key Share Entry: Group: Unknown (4588), Key Exchange length: 1216
      Key Share Entry: Group: x25519, Key Exchange length: 32
```

22. At this point, does the client know that this DH curve is supported by the server?

The client does not know that DH is supported by the server, but the client guesses and hopes that the server will agree, thus saving us from exchanging keys.

Server Hello:

23. What version exists in the Record Header? What version exists in the Server Hello Version field?

```
Transport Layer Security
  TLSv1.3 Record Layer: Handshake Protocol: Server Hello
    Content Type: Handshake (22)
    Version: TLS 1.2 (0x0303)
    Length: 122
    Handshake Protocol: Server Hello
      Handshake Type: Server Hello (2)
      Length: 118
      Version: TLS 1.2 (0x0303)
      Random: e5179ae02af9b3fb65a84b368bc4316173a4ad6e34
      Session ID Length: 32
      Session ID: 39d56ae08caa00402e8e73c52857faa0ac525e
      Cipher Suite: TLS_AES_256_GCM_SHA384 (0x1302)
      Compression Method: null (0)
      Extensions Length: 46
      Extension: supported_versions (len=2)
        Type: supported_versions (43)
        Length: 2
        Supported Version: TLS 1.3 (0x0304)
```

24. What are the first 6 hex digits of the Server Random number?

```
Transport Layer Security
  TLSv1.3 Record Layer: Handshake Protocol: Server Hello
    Content Type: Handshake (22)
    Version: TLS 1.2 (0x0303)
    Length: 122
    Handshake Protocol: Server Hello
      Handshake Type: Server Hello (2)
      Length: 118
      Version: TLS 1.2 (0x0303)
      Random: e5179ae02af9b3fb65a84b368bc4316173a4ad6e34
      Session ID Length: 32
      Session ID: 39d56ae08caa00402e8e73c52857faa0ac525e
      Cipher Suite: TLS_AES_256_GCM_SHA384 (0x1302)
      Compression Method: null (0)
      Extensions Length: 46
      Extension: supported_versions (len=2)
        Type: supported_versions (43)
        Length: 2
        Supported Version: TLS 1.3 (0x0304)
```

25. What are the first 6 hex digits of the Session ID? How does it compare to the client Session ID?

Why? Same as client Session ID

```
Transport Layer Security
  TLSv1.3 Record Layer: Handshake Protocol: Server Hello
    Content Type: Handshake (22)
    Version: TLS 1.2 (0x0303)
    Length: 122
    Handshake Protocol: Server Hello
      Handshake Type: Server Hello (2)
      Length: 118
      Version: TLS 1.2 (0x0303)
      Random: e5179ae02af9b3fb65a84b368bc4316173a4ad6e34
      Session ID Length: 32
      Session ID: 39d56ae08caa00402e8e73c52857faa0ac525e
      Cipher Suite: TLS_AES_256_GCM_SHA384 (0x1302)
      Compression Method: null (0)
      Extensions Length: 46
      Extension: supported_versions (len=2)
        Type: supported_versions (43)
        Length: 2
        Supported Version: TLS 1.3 (0x0304)
```

26. Which Cipher Suites did the server choose?

```
Handshake Protocol: Server Hello
    Handshake Type: Server Hello (2)
    Length: 118
    Version: TLS 1.2 (0x0303)
    Random: e5179ae02af9b3fb65a84b368bc4316173a4ad6e344d7e0e…
    Session ID Length: 32
    Session ID: 39d56ae08caa00402e8e73c52857faa0ac525e2be0e4cf70…
    Cipher Suite: TLS_AES_256_GCM_SHA384 (0x1302)
```

27. Use the Extension "Key_Share". Which Diffie-Hellman curve is chosen by the server?

```
Extension: key_share (len=36)
    Type: key_share (51)
    Length: 36
    Key Share extension
        Key Share Entry: Group: x25519, Key Exchange length: 32
```

Now the client and server have completed the coordination of exchanging the shared secret in DH.

28. Did the client guess correctly? What would happen if not?

Yes, the client guessed correctly, the server supports what it suggested. And if the client did not guess correctly, then the server sends HelloRetryRequest the HASH of this as the RANDOM

<u>Certificate</u>:

Decrypt the TLS communication. There are 3 subfields under "Certificates". A certificate, an "Extension" and another certificate

29. Who holds the first certificate? Look into the "subject" field

Click on the certificate packet **-> subject: jct.ac.il** [subject = למי התעודה שייכת]

```
Certificates (3079 bytes)
    Certificate Length: 1267
    Certificate […]: 308204ef308203d7a003020102021203 5d0d760faeeaf392607c91cd3fd560337e300d06092a864886f70d01010b0500
        signedCertificate
            version: v3 (2)
            serialNumber: 0x035d0d760faeeaf392607c91cd3fd560337e
            signature (sha256WithRSAEncryption)
            issuer: rdnSequence (0)
                rdnSequence: 3 items (id-at-commonName=R11,id-at-organizationName=Let's Encrypt,id-at-countryName=US)
                    RDNSequence item: 1 item (id-at-countryName=US)
                        RelativeDistinguishedName item (id-at-countryName=US)
                    RDNSequence item: 1 item (id-at-organizationName=Let's Encrypt)
                        RelativeDistinguishedName item (id-at-organizationName=Let's Encrypt)
                    RDNSequence item: 1 item (id-at-commonName=R11)
                        RelativeDistinguishedName item (id-at-commonName=R11)
            validity
            subject: rdnSequence (0)
                rdnSequence: 1 item (id-at-commonName=jct.ac.il)
                    RDNSequence item: 1 item (id-at-commonName=jct.ac.il)
                        RelativeDistinguishedName item (id-at-commonName=jct.ac.il)
```

30. Who issued the first certificate? Look into the "issuer" field

Click on the certificate packet -> **issuer: R11 Lets Encrypt** [issuer = מי שחתם על התעודה]

31. What is the purpose of the "Extension"? Which protocol is used?

Extension means that the certificate is valid and the protocol used is OCSP.

```
Certificates (3079 bytes)
    Certificate Length: 1267
    Certificate [...]: 308204ef308203d7a003020102021203
    Extensions Length: 512
    Extension: status_request (len=508)
        Type: status_request (5)
        Length: 508
        Certificate Status Type: OCSP (1)
        OCSP Response Length: 504
        OCSP Response
```

```
Certificate [...]: 308204ef308203d7a0030201020212035d0d760faeeaf3926
Extensions Length: 512
Extension: status_request (len=508)
    Type: status_request (5)
    Length: 508
    Certificate Status Type: OCSP (1)
    OCSP Response Length: 504
    OCSP Response
        responseStatus: successful (0)
        responseBytes
            ResponseType Id: 1.3.6.1.5.5.7.48.1.1 (id-pkix-ocsp-basic)
            BasicOCSPResponse
                tbsResponseData
                    responderID: byName (1)
                        byName: 0
                    producedAt: Dec 14, 2024 01:54:00.000000000 Jerusale
                    responses: 1 item
                        SingleResponse
                            certID
                                hashAlgorithm (SHA-1)
                                issuerNameHash: 1a52b9b459e4c33398217e866d4b
                                issuerKeyHash: 08b9113ba5d08518b4ea0fa0ad9f8
                                serialNumber: 0x035d0d760faeeaf392607c91cd3
                            certStatus: good (0)
                            thisUpdate: Dec 14, 2024 01:54:00.000000000 Je
                            nextUpdate: Dec 21, 2024 01:53:58.000000000 Je
```

certificate: good – זה מעיד שהתעודה טובה כלומר בתוקף

32. Who holds the second certificate? Look into the “subject” field

    R11 Lets Encrypt

33. Who issued the second certificate? Look into the “issuer” field

    ISRG Root x1 – (זה כבר נמצא על המחשב שלנו)

The End, Good luck!